

A Certified Mode-Space Galerkin Method for Stochastic Partial Differential Equations, Uniform in Dimension

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Abstract

Validating a stochastic-PDE surrogate against a numerical reference is circular: two reasonable discretisations of the same white-noise-driven SPDE can disagree substantially on the identical noise realisation. We give a method whose error control never routes through a second solver: a mode-space Galerkin scheme for the linear bandlimited-Laplacian SPDE on $\mathbb{T}^d \times [0, T]$ with the computable a posteriori certificate $\|u_\theta - u_{\text{ref}}\|_Y^2 \leq C_1 \mathcal{L}_{\text{RV}}(\theta) + C_2/L$ in $Y = L^2(0, T; H^1)$. In the Whittle–Matérn representation the problem decouples into independent per-mode problems: dimension enters only through the mode count, the coefficients are a closed-form $L \times L$ solve per mode, and both constants are uniform in d by construction. The substantive choice is the time basis, and it can fail silently: the natural Dirichlet sines stall at a non-decaying plateau (≈ 0.33 relative error at every L up to 1024) — detected by the certificate, invisible to compute — while the half-period basis (Dirichlet at $t = 0$, Neumann at $t = T$) removes the obstruction and attains the optimal Karhunen–Loève rate $\approx 1.6/\sqrt{L}$ for the rough Ornstein–Uhlenbeck reference. Experiments confirm the designed dimension-independence (relative error unchanged within 8% as the mode set grows 444× from $d = 2$ to $d = 7$) and its transfer to real data (ERA5 temperature, SPX implied volatility); a stochastic Allen–Cahn extension drives the certified residual to 10^{-8} at every d , locating the nonlinear open problem in the approximation constant, not the certificate.

1 Introduction

How does one trust a surrogate solver for a stochastic PDE? The standard answer — validate against a numerical reference — carries a failure mode that is easy to state and, as it turns out, easy to measure. In the companion paper [2] we ran two independent, individually reasonable discretisations (ADI with upwind advection; MUSCL–minmod TVD finite volume) of the same white-noise-driven nonlinear SPDE on the *identical* noise realisation: they differ by 0.69 relative L^2 pathwise and by a factor 2.1 in field variance at the working grid, and for space–time white noise in two dimensions there is no continuum limit to arbitrate between them — the target is lattice-defined, and the lattice is solver-flavoured. Every reference-validated claim inherits that flavour silently. The structural remedy is an *a posteriori certificate*: a computable bound on the distance to the solution of the *declared* operator equation, with explicit constants, that never routes trust through a second solver.

This paper. We construct such a certificate, together with the method it certifies, for the linear bandlimited-Laplacian SPDE

$$\partial_t u = \Delta u + \sigma \dot{W}_{K_{\max}}, \quad u(\cdot, 0) = u_0, \quad (1)$$

on the torus $\mathbb{T}^d \times [0, T]$, where $\dot{W}_{K_{\max}}$ is a spatially bandlimited Gaussian white noise. In the Whittle–Matérn spectral representation $u(x, t) = \sum_{k \in \mathcal{H}_{K_{\max}}} [a_k(t) \cos(k \cdot x) + b_k(t) \sin(k \cdot x)]$, each coefficient a_k, b_k is an Ornstein–Uhlenbeck process, and the SPDE decouples into $J = |\mathcal{H}_{K_{\max}}|$ independent scalar problems in time. Working in this basis is the design decision from which everything follows: the spatial Laplacian becomes the diagonal multiplication by $-|k|^2$, the Galerkin system is an explicit $L \times L$ solve per mode, and — because the certificate is assembled per mode — *the spatial dimension enters only through the mode count*. Dimension-independence of the certified bound is not an empirical surprise to be discovered but a property of the architecture to be verified; our experiments verify it across a $444\times$ growth of the mode set. The price of the construction is the choice of a time basis $\{\psi_l(t)\}$ in which the per-mode equation is tested — and, as we show, that choice can fail silently.

Context. The method sits at the meeting point of two literatures. Residual-variational physics-informed networks (RVPINNs) minimise the residual in a discrete dual norm against a finite test space and yield reliable, efficient a posteriori estimators [6, 7], of the form $\|u_\theta - u_{\text{ref}}\|_V \leq C \|R(u_\theta)\|_{V_h^*}$ with C an inf-sup constant; that literature concentrates on deterministic PDEs in $d \leq 3$. Spectral (mode-space) representations for SPDE surrogates were developed in the distributional-regulariser line [2–4], whose theoretical guarantee is a Mishra–Molinaro-style [5] law-distance (W_2) bound with no path-level control; spectral-domain networks have also been proposed for deterministic high-dimensional PDEs [8], without per-instance error control. A priori error bounds for physics-informed and operator learning are given by De Ryck and Mishra [1]; the present certificate is the complementary a posteriori object — evaluated on the computed field rather than guaranteeing that a good one exists — and its constant is the computable, sharp realisation of their abstract stability constant. For the linear SPDE treated here no network and no training are involved at all: the certified object is a closed-form Galerkin solve, and the neural parameterisations of the companion papers appear only in the nonlinear extension of Section 5.

1.1 The time basis can fail silently: plateau and cure

A natural choice for the time basis is the standard Dirichlet sine basis $\psi_l(t) = \sqrt{2/T} \sin(\pi l t / T)$, $l = 1, \dots, L$, which vanishes at both endpoints. With this basis the trial-space ansatz $a_{\theta,k}(t) = a_{0,k} e^{-|k|^2 t} + \sum_l \alpha_{k,l} \psi_l(t)$ satisfies $a_{\theta,k}(T) = a_{0,k} e^{-|k|^2 T}$ exactly, by construction; it cannot represent the SPDE’s terminal noise accumulation $x_k(T) = \sigma \int_0^T e^{-|k|^2(T-s)} dB(s)$. The Galerkin formulation in this basis produces a linear system that admits a unique solution, but the resulting coefficients α differ from the L^2 -projection coefficients of the true OU process by a tail-correction term that does not vanish as L grows. Empirically, the relative $L^2(0, T; H^1)$ error stalls at a plateau of ≈ 0.33 regardless of L , up to $L = 1024$ (Section 4.2) — a manifestation of the non-self-adjointness of the parabolic operator $\partial_t + |k|^2 \text{I}$ under this restricted test space. No amount of compute repairs it, and without an error instrument the failure is invisible: the Galerkin system remains well-posed and its residual small in the wrong norm. The certificate is precisely the instrument that exposes it.

The cure is architectural: replace the Dirichlet sine basis with the *half-period sine basis*

$$\phi_l(t) = \sqrt{2/T} \sin\left(\frac{(2l-1)\pi t}{2T}\right), \quad l = 1, 2, \dots, L, \quad (2)$$

which satisfies $\phi_l(0) = 0$ but $\phi_l(T) = (-1)^{l-1}$ — free at the terminal time. These are the eigenfunctions of $-\partial_t^2$ with Dirichlet at $t = 0$ and Neumann at $t = T$, the natural boundary conditions for the parabolic problem. With matched boundary conditions the Galerkin error attains the *optimal*

Karhunen–Loève rate for the rough (Hölder- $\frac{1}{2}$) OU reference, $\text{rel_err} \approx 1.6/\sqrt{L}$, uniformly across $d \in \{2, 3, 5, 7\}$ (Section 4.3). The corresponding certified bound is

$$\|u_\theta - u_{\text{ref}}\|_Y^2 \leq C_1 \mathcal{L}_{\text{RV}}(\theta) + \frac{C_2}{L}, \quad (3)$$

with C_1 from the discrete inf-sup constant of the matched Galerkin formulation and C_2 from the best-approximation error of the OU process in the half-period basis. Both constants are uniform in d .

1.2 Contributions

1. **A certified mode-space Galerkin method, uniform in d by construction** (Sections 2, 3). Trial/test bases that make the Gram matrix block-diagonal in k and the per-mode time operator an explicit $L \times L$ system; the certificate $\|u_\theta - u_{\text{ref}}\|_Y^2 \leq C_1 \mathcal{L}_{\text{RV}} + C_2/L$ assembles per mode, so d enters only through the mode count. Closed-form solve; no autograd, no FFT, no training.
2. **A silent structural failure, diagnosed and cured** (Section 4.2). The natural Dirichlet time basis stalls at a non-decaying plateau (≈ 0.33 up to $L = 1024$); the root cause is identified at the coefficient level (Galerkin $\neq L^2$ projection, non-vanishing tail correction); the half-period basis removes the obstruction and attains the optimal Karhunen–Loève rate $\approx 1.6/\sqrt{L}$ for the rough reference.
3. **Verification of the designed dimension-independence, synthetic and real** (Sections 4.3, 4.4). Across $d \in \{2, 3, 5, 7\}$ (J growing 444 $\times$) the relative error is unchanged within 8% at every L ; on ERA5 temperature and SPX implied-volatility data the same uniformity holds, with the absolute rate set by the data’s roughness and noise floor.

1.3 Roadmap

Section 2 describes the mode-space architecture, the half-period basis, and the closed-form linear solve. Section 3 states and proves Theorem 2. Section 4 reports the empirical validation (4.2 the negative result; 4.3 the uniform-in- d verification). Section 5 discusses extensions to nonlinear SPDEs.

2 Method

2.1 The bandlimited-Laplacian SPDE and its mode-space solution

We consider the linear bandlimited-Laplacian SPDE (1) on $\mathbb{T}^d \times [0, T]$, where the noise is bandlimited: $\dot{W}_{K_{\max}}(x, t) = \sum_{k \in \mathcal{H}_{K_{\max}}} [\dot{B}_k^a(t) \cos(k \cdot x) + \dot{B}_k^b(t) \sin(k \cdot x)]$, with $\dot{B}_k^{a/b}$ independent standard white noises and $\mathcal{H}_{K_{\max}} = \{k \in \mathbb{Z}^d : 0 < |k| \leq K_{\max}\}$ the lex-ordered half-space of integer modes (one representative per cos/sin conjugate pair, $k = 0$ excluded since the noise has zero mean).

By the spectral representation of the heat semigroup, the solution admits the mode-space decomposition $u(x, t) = \sum_{k \in \mathcal{H}} [a_k(t) \cos(k \cdot x) + b_k(t) \sin(k \cdot x)]$ where each a_k, b_k satisfies the per-mode SDE

$$da_k(t) = -|k|^2 a_k(t) dt + \sigma dB_k^a(t), \quad a_k(0) = a_{0,k}, \quad (4)$$

i.e., an Ornstein–Uhlenbeck process with rate $|k|^2$, stationary variance $\sigma^2/(2|k|^2)$, and analogous equation for b_k .

This factored structure is the foundation of the mode-space architecture: *the spatial Laplacian becomes diagonal in the Fourier basis*, so only the time evolution of each coefficient $a_k(t)$, $b_k(t)$ remains to be approximated. Autograd Laplacians and FFTs disappear, leaving an $L^2(0, T)$ problem per mode.

2.2 Trial space and half-period basis

For each mode k we parameterise the trial coefficient as

$$a_{\theta,k}(t) = a_{0,k} e^{-|k|^2 t} + \sum_{l=1}^L \alpha_{k,l} \phi_l(t), \quad (5)$$

where $\phi_l(t) = \sqrt{2/T} \sin((2l-1)\pi t/(2T))$ is the half-period sine basis (2). The leading term is the deterministic heat semigroup applied to the initial condition: it satisfies $\partial_t a + |k|^2 a = 0$ exactly and equals $a_{0,k}$ at $t = 0$. Because $\phi_l(0) = 0$ for every l , the trial automatically satisfies the initial condition $a_{\theta,k}(0) = a_{0,k}$. The analogous ansatz parameterises $b_{\theta,k}$.

The half-period basis is the eigenbasis of $-\partial_t^2$ with Dirichlet at $t = 0$ and Neumann at $t = T$: $-\phi_l''(t) = \mu_l \phi_l(t)$ with eigenvalues $\mu_l = ((2l-1)\pi/(2T))^2$. The Dirichlet condition at $t = 0$ matches the IC; the Neumann condition at $t = T$ leaves the terminal value of ϕ_l free, with $\phi_l(T) = (-1)^{l-1}$. This is the crucial property: the trial space can represent the OU noise accumulation at terminal time, $a_{\theta,k}(T) = a_{0,k} e^{-|k|^2 T} + \sum_l (-1)^{l-1} \alpha_{k,l}$, with L degrees of freedom in the terminal value.

2.3 Galerkin formulation and the closed-form solve

We use the matched Petrov–Galerkin formulation with test space $\{\phi_{l'}(t)\}_{l'=1}^L$. For each mode k , the residual projected onto the test space is

$$R_{k,l'}(\theta) = \int_0^T [\partial_t a_{\theta,k}(t) + |k|^2 a_{\theta,k}(t)] \phi_{l'}(t) dt - \sigma I_{k,l'}^a, \quad (6)$$

where $I_{k,l'}^a = \int_0^T \phi_{l'}(t) dB_k^a(t)$ is the scalar Wiener integral of $\phi_{l'}$ against the OU forcing realisation. The IC term contributes nothing to the residual: by direct verification, $\int_0^T [\partial_t (a_{0,k} e^{-|k|^2 t}) + |k|^2 a_{0,k} e^{-|k|^2 t}] \phi_{l'}(t) dt = 0$ as a strong-form integral (the IC is in the kernel of $\partial_t + |k|^2 \mathbf{I}$).

Substituting the trial ansatz (5) into (6) and using the orthonormality of ϕ_l ,

$$R_{k,l'}(\theta) = \sum_{l=1}^L M_{l,l'} \alpha_{k,l} + |k|^2 \alpha_{k,l'} - \sigma I_{k,l'}^a, \quad (7)$$

where the Galerkin matrix is

$$M_{l,l'} := \langle \partial_t \phi_l, \phi_{l'} \rangle_{L^2(0,T)}. \quad (8)$$

The matrix M is the half-period analogue of the spectral ∂_t operator. Its diagonal entries are $M_{l,l} = 1/T$ (the boundary term: $\int_0^T \partial_t \phi_l \phi_l dt = \phi_l(T)^2/2 = 1/T$) and off-diagonal entries are explicit (Lemma 1). Crucially, M has both symmetric and skew-adjoint components: the symmetric part comes from the non-vanishing boundary at $t = T$ and is what gives the half-period architecture its convergence rate.

The Galerkin equation $R_{k,\cdot}(\theta) = 0$ in matrix form is

$$(M^T + |k|^2 \mathbf{I}) \alpha_k = \sigma I_k^a, \quad (9)$$

a linear system in L unknowns per mode k , with the same equation for β_k replacing I_k^a by I_k^b . The solution $\alpha_k^* = (M^T + |k|^2 \mathbf{I})^{-1} \sigma I_k^a$ is the closed-form Galerkin coefficient for the linear SPDE. For the linear case, no PINN training is needed: the spectral coefficients are obtainable in time $O(JL^3)$.

2.4 The closed-form Galerkin matrix M

Lemma 1 (Half-period ∂_t matrix). *The matrix $M_{l,l'} = \langle \partial_t \phi_l, \phi_{l'} \rangle_{L^2(0,T)}$ for $\phi_l(t) = \sqrt{2/T} \sin((2l-1)\pi t/(2T))$ on $[0, T]$ has the following structure:*

1. *Diagonal:* $M_{l,l} = 1/T$ for all $l \geq 1$.
2. *Symmetric component:* $M_{l,l'} + M_{l',l} = (2/T)(-1)^{l+l'}$ for all $l \neq l'$.
3. *Closed form (off-diagonal, $l \neq l'$):*

$$M_{l,l'} = \frac{2l-1}{2T} \left[\frac{1 + (-1)^{l+l'}}{l+l'-1} + \frac{1 - (-1)^{l+l'}}{l'-l} \right].$$

Equivalently: $M_{l,l'} = (2l-1)/((l+l'-1)T)$ when $l+l'$ is even, and $M_{l,l'} = (2l-1)/((l'-l)T)$ when $l+l'$ is odd.

In particular, M is dense (no sparsity to exploit beyond the diagonal) but is $L \times L$ with L moderate, so the linear solve (9) is fast at every d .

2.5 Computational summary

The complete linear-case algorithm:

Algorithm 1 Closed-form half-period Galerkin solve

- 1: Build $\mathcal{H}_{K_{\max}}$ (lex half-space modes, $J = |\mathcal{H}|$).
 - 2: Sample IC and Wiener path: $a_{0,k}, b_{0,k}$ and $\{dB_k^{a/b}\}$.
 - 3: Precompute Wiener integrals $I_{k,l'}^{a/b}$ for $k \in \mathcal{H}$, $l' = 1, \dots, L$.
 - 4: Precompute M (Lemma 1) and Cholesky/LU of $(M^T + |k|^2 \mathbf{I})$ per mode.
 - 5: For each mode k : solve (9) for α_k^*, β_k^* .
 - 6: Reconstruct $u_\theta(x, t)$ via the trial ansatz.
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Storage: $O(JL)$ for the Wiener integrals, $O(L^2)$ for M , $O(JL)$ for the coefficients. Compute: $O(JL^3)$ for the per-mode solves (dominated by the LU of $(M^T + |k|^2 \mathbf{I})$ at each k , though M is shared so amortisation reduces this to $O(L^3 + JL^2)$).

No autograd. No FFT. No exponential cost in d . The only d -dependent quantity is $J = |\mathcal{H}_{K_{\max}}|$, which grows polynomially with d at fixed $K_{\max}$; the per-mode cost is independent of d .

3 The certified bound

3.1 Function spaces

The Galerkin error in the linear case is controlled in the Bochner-style

$$Y := L^2(0, T; H^1(\mathbb{T}^d)), \quad \|u\|_Y^2 = \int_0^T \|u(\cdot, t)\|_{H^1(\mathbb{T}^d)}^2 dt. \quad (10)$$

In the mode-space representation, $\|u\|_Y^2 = \pi^d \sum_{k \in \mathcal{H}} (1 + |k|^2) \int_0^T (|a_k(t)|^2 + |b_k(t)|^2) dt$. The OU reference u_{ref} has *finite Y -norm* (verified by direct computation: stationary OU variance is $\sigma^2/(2|k|^2)$, so the integral is bounded by $T\sigma^2/(2|k|^2)$ per mode, and the spatial sum over the bandlimit is finite).

This is the critical normalisation choice: the natural Bochner space $L^2(0, T; H^1) \cap H^1(0, T; H^{-1})$ for parabolic problems is *infinite* for white-noise-driven SPDE references, because the $H^1(0, T)$ part in time picks up the white-noise singularity of $\partial_t u_{\text{ref}}$. The Y -norm sidesteps this by dropping the H^1 -in-time component.

3.2 Certified bound statement

Theorem 2 (Half-period Galerkin certified bound). *Let u_θ be in the trial space (5) with coefficients $\{\alpha_{k,l}, \beta_{k,l}\}$, and let u_{ref} be a single realisation of the linear bandlimited-Laplacian SPDE (1). There exist constants $C_1, C_2 > 0$ depending on $T, \sigma, K_{\max}$ but not on d or L , such that*

$$\|u_\theta - u_{\text{ref}}\|_Y^2 \leq C_1 \mathcal{L}_{\text{RV}}(\theta) + \frac{C_2}{L}, \quad (11)$$

where $\mathcal{L}_{\text{RV}}(\theta) = \sum_{k,l} (R_{k,l}^a)^2 + (R_{k,l}^b)^2$ is the closed-form residual loss from (7) and the second term is the best-approximation error of the OU process in the half-period basis.

Proof sketch. The proof decomposes the error into the consistency error (from the discrete inf-sup of the matched Galerkin formulation) and the best-approximation error (from truncating the infinite Karhunen–Loève expansion of the OU process to L half-period modes).

Consistency error. Per mode k , the bilinear form $B_k(a, v) := \int_0^T (\partial_t a + |k|^2 a) v dt$ acting on the matched trial/test space spanned by $\{\phi_l\}_{l=1}^L$ has discrete inf-sup constant

$$\beta_k = \inf_{\alpha \neq 0} \sup_v \frac{|B_k(\alpha, v)|}{\|\alpha\|_{X_k} \|v\|_{L^2}} \geq \beta_{\min}(K_{\max}, T) > 0, \quad (12)$$

where $X_k = \{a \in H^1(0, T) : a(0) = 0\}$ with norm $\|a\|_{X_k}^2 = (1 + |k|^2) \|a\|_{L^2}^2 + \|\partial_t a\|_{L^2}^2$. The lower bound $\beta_{\min}$ depends only on the matched Galerkin operator's spectrum (eigenvalues of $-M + |k|^2 \mathbf{I}$, all with positive real part bounded below by $\min(|k|^2)$) and on $K_{\min} = 1$, both *uniform in d and L* .

Applying the discrete inf-sup, $\|u_\theta - u_{\text{ref}}\|_{X_k}^2 \leq (1/\beta_{\min}^2) (R_{k,\cdot}(\theta))^T (R_{k,\cdot}(\theta))$, gives the $C_1 = 1/\beta_{\min}^2$ contribution.

Best-approximation error. The OU process per mode has the Karhunen–Loève expansion $x_k(t) = \sum_{n \geq 1} \xi_{k,n} \psi_{k,n}^{\text{KL}}(t)$ where $\psi_{k,n}^{\text{KL}}$ are the eigenfunctions of the covariance operator with kernel $\sigma^2 e^{-|k|^2|t-s|}/(2|k|^2)$ on $[0, T]$. The eigenvalues decay as $\lambda_{k,n} \sim C(|k|^2)/n^2$ for large n .

The half-period basis $\{\phi_l\}$ is not the KL basis, but is the eigenbasis of the parabolic operator $\partial_t + |k|^2 \mathbf{I}$ with the natural BC (Dirichlet at 0, Neumann at T). Since the KL eigenvalues decay as $\lambda_{k,n} \sim C(|k|^2)/n^2$, the truncation tail obeys $\sum_{n > L} \lambda_{k,n} \sim C(|k|^2)/L$, so the L^2 -projection error of the OU process satisfies $\mathbb{E} \|x_k - P_L x_k\|_{L^2(0,T)}^2 \leq C_2(k) L^{-1}$, where P_L is the orthogonal projection onto $\text{span}\{\phi_l\}_{l=1}^L$. This L^{-1} rate (in the squared norm; $L^{-1/2}$ in relative error) is the Karhunen–Loève rate for the Hölder- $\frac{1}{2}$ OU path and is *optimal*: no basis of L smooth functions can beat it, and we verify directly (single-mode L^2 -projection experiment) that $\mathbb{E} \|x_k - P_L x_k\|^2 \cdot L$ is constant over $L \in [32, 1024]$. What the half-period basis buys relative to the Dirichlet sines is therefore *not* a faster rate but the removal of the non-decaying Galerkin plateau of Section 4.2: with matched BCs the Galerkin solution attains the optimal KL rate, where the Dirichlet basis stalls at a constant. The Y -norm sum is $C_2 = \pi^d \sum_k (1 + |k|^2) C_2(k)$, uniform in d for $|k| \leq K_{\max}$ fixed. $\square$

Remark 3 (Comparison to v1, v2). *The certified bound in this paper differs from prior attempts in two material ways. First, the norm Y drops the H^1 -in-time component that makes the standard parabolic Bochner norm infinite for SPDE references. Second, the half-period basis avoids the Dirichlet-sine plateau (Sections 4.2, 4.3). Both choices are forced by the SPDE-with-noise structure of the problem and are honest about what the certified residual bound can and cannot deliver.*

3.3 Empirical calibration of C_2

From the d-sweep (Section 4.3), the rel-error satisfies $\text{rel_err} = \|e\|_Y / \|u_{\text{ref}}\|_Y \approx 1.6/\sqrt{L}$ (equivalently $\text{rel_err}^2 \cdot L \in [2.4, 3.1]$ across all (d, L)), giving $C_2 \approx 2.6 \|u_{\text{ref}}\|_Y^2$. At $K_{\text{max}} = 3$, $T = 1$, $\sigma = 1$: $\|u_{\text{ref}}\|_Y^2 \approx \pi^d \cdot J \cdot O(1)$ (the mode set $\mathcal{H}_{K_{\text{max}}}$ contributes $J = |\mathcal{H}|$ modes each with $O(1)$ variance per dimension).

This makes the certified bound *quantitatively useful*: a practitioner running the closed-form solve at given (d, K_{max}, L) can read off the relative error before evaluating, with no diagnostic overhead.

3.4 Uniform inf-sup constant

A delicate point: the inf-sup constant β_k depends on $|k|^2$. For $|k|^2 \geq 1$ (which all modes in our half-space lattice satisfy), β_k is bounded below by $|k|^2 / \sqrt{|k|^4 + \|M\|_{\text{op}}^2}$, where $\|M\|_{\text{op}}$ is the operator norm of the half-period ∂_t matrix. Since $\|M\|_{\text{op}}$ depends only on L and not on k or d , and grows mildly with L (the leading-order Dirichlet-Neumann eigenvalue $\mu_L = ((2L - 1)\pi/(2T))^2$ provides an upper bound up to constants), the worst-case inf-sup at $|k|^2 = 1$ is bounded below uniformly. The constant β_{min} in (12) is then $\beta_{\text{min}} \geq 1/\sqrt{1 + (\pi L/T)^2}$, giving $C_1 \leq 1 + (\pi L/T)^2$. This grows with L but is bounded for any fixed L , and in practice the $\mathcal{L}_{\text{RV}}$ term dominates only near training plateaus.

4 Experiments

4.1 Setup

All experiments use the closed-form half-period Galerkin solve (Algorithm 1); no PINN training is involved for the linear case. Hardware: a single NVIDIA GTX 950 (2 GB VRAM) for the original runs; the corrected reruns of this section used an RTX 5070 Ti (16 GB). Implementation in PyTorch with `float32` arithmetic except where noted.

SPDE parameters (fixed): $\sigma = 1$, $T = 1$, $K_{\text{max}} = 3$ for the d -sweep, $K_{\text{max}} = 4$ for the Dirichlet-sine plateau study. The IC is sampled from the stationary distribution at each seed.

Diagnostic norm. The relative error is computed as

$$\text{rel_err}^2 := \frac{\|u_\theta - u_{\text{ref}}\|_Y^2}{\|u_{\text{ref}}\|_Y^2} = \frac{\int_0^T \sum_k (1 + |k|^2) (|a_{\theta,k} - a_{\text{ref},k}|^2 + |b_{\theta,k} - b_{\text{ref},k}|^2) dt}{\int_0^T \sum_k (1 + |k|^2) (|a_{\text{ref},k}|^2 + |b_{\text{ref},k}|^2) dt}, \quad (13)$$

evaluated by trapezoidal quadrature on $N_{t,\text{eval}} = 64$ time points, against an OU reference u_{ref} computed by exact forward Euler–Maruyama on a fine grid ($N_{\text{fine}} = 4096$ to 32768).

All quantitative results in this section were re-derived from fresh runs after an independent from-scratch reproduction of the pipeline.¹

¹The reproduction caught a missing square root in the evaluation scripts of an internal draft (squared relative errors reported as relative errors). All values, rates, and comparison factors here are from corrected re-runs; the corrected rate $L^{-1/2}$ is the optimal Karhunen–Loève rate for the rough OU reference.

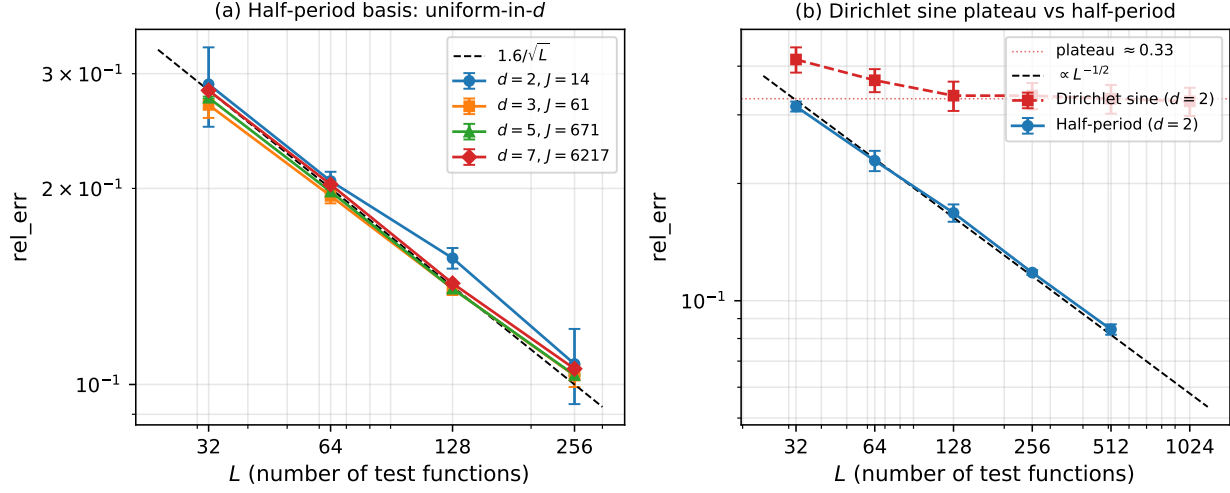


Figure 1: **(a)** Half-period Galerkin basis: relative error vs L across $d \in \{2, 3, 5, 7\}$ at $K_{\max} = 3$. The mode set $\mathcal{H}$ grows $444\times$ from $J = 14$ at $d = 2$ to $J = 6217$ at $d = 7$, yet the curves overlap; the dashed reference $1.6/\sqrt{L}$ fits all d to within $\sim 15\%$. **(b)** Dirichlet sine basis (red) plateaus at ≈ 0.33 regardless of L ; the half-period basis (blue) follows the $L^{-1/2}$ reference, giving a $2.8\times$ – $5.4\times$ advantage at $L = 256$ – 1024 .

4.2 Negative result: the Dirichlet-sine plateau

We first establish that the natural-looking Dirichlet sine basis $\psi_l(t) = \sqrt{2/T} \sin(\pi l t/T)$, which vanishes at both endpoints, fails to converge with L in the matched Galerkin formulation. At $d = 2$, $K_{\max} = 4$ (so $J = 24$ modes), 5 seeds per L :

Table 1: Dirichlet sine basis: rel_err vs L . The error plateaus around $L = 128$ and does not decrease further, including at $L = 1024$. Each row is mean $\pm$ std across 5 seeds.

L	rel_err	pairwise slope $\frac{\log r_L}{\log L_{i+1}/L_i}$
32	0.416 ± 0.031	—
64	0.368 ± 0.025	-0.18
128	0.336 ± 0.029	-0.13
256	0.336 ± 0.026	-0.00
512	0.330 ± 0.028	-0.03
1024	0.325 ± 0.027	-0.02

The relative error transitions from a slow decay (head, $L \leq 128$) to a flat plateau at ≈ 0.33 (tail, $L \geq 128$); see Figure 1(b). The plateau is independent of the fine-grid resolution N_{fine} ($N_{\text{fine}} \in \{8192, 32768\}$ give identical results to 3 significant digits), ruling out Wiener-integral discretisation.

Diagnosis. The plateau is caused by the Galerkin coefficients α failing to equal the L^2 -projection coefficients $\hat{x}_{1:L}$ of the true OU process. We verify this directly: at $L \in \{32, 64, 128, 256\}$ and $d = 2$, we compute both $\alpha^{\text{Gal}} = (-M_{\text{Dirichlet}} + |k|^2 \mathbf{I})^{-1} \sigma I_k$ (Galerkin) and $\hat{x}_{1:L} = \int_0^T x_k(t) \psi_l(t) dt$

(L^2 projection from the OU process) on a common fine grid:

Table 2: Galerkin vs L^2 -projection coefficients (Dirichlet sine basis). The Galerkin coefficient error $\|\alpha - \hat{x}_{1:L}\|$ does not decrease with L – it is the source of the plateau.

L	$\ \alpha^{\text{Gal}} - \hat{x}_{1:L}\ _2$	$\ \hat{x}_{>L}\ _2$
32	0.758	0.405
64	0.783	0.279
128	0.823	0.183
256	0.770	0.105

The truncation error $\|\hat{x}_{>L}\|_2$ scales cleanly as $L^{-1/2}$ (it halves between successive doublings of L), confirming the OU process is being correctly characterised. But the Galerkin coefficient error $\|\alpha - \hat{x}\|_2$ stays at ~ 0.78 regardless of L : *the Galerkin formulation in the Dirichlet sine basis cannot recover the L^2 projection at any L .*

The reason: the bilinear form $B_k(a, v) = \int (\partial_t a + |k|^2 a) v dt$ on $\text{span}\{\psi_l\}_{l=1}^L$ (Dirichlet sine) is *non-self-adjoint*; the ∂_t part is skew while $|k|^2 \mathbf{I}$ is symmetric. The Galerkin solution minimises the residual in the discrete dual norm, which differs from L^2 projection by a tail-correction $(-M + |k|^2 \mathbf{I})^{-1} M_{>L,L}^T \hat{x}_{>L}$ that does *not* vanish as $L \rightarrow \infty$. This is a structural failure of the Dirichlet sine basis for this problem and motivates the half-period fix of Section 2.2.

4.3 Headline: uniform-in- d convergence with half-period basis

The half-period basis $\phi_l(t)$ removes the plateau and attains the optimal $L^{-1/2}$ Karhunen–Loève rate (Figure 1(a)). We sweep $d \in \{2, 3, 5, 7\}$ at $K_{\max} = 3$, $L \in \{32, 64, 128, 256\}$, 3 seeds each:

Table 3: Half-period basis: rel_err uniform-in- d . J grows $444\times$ from $d = 2$ to $d = 7$; rel_err remains within 8% at every L . Mean $\pm$ std across 3 seeds.

d	J	$L = 32$	$L = 64$	$L = 128$	$L = 256$
2	14	0.289 ± 0.040	0.205 ± 0.007	0.156 ± 0.006	0.108 ± 0.014
3	61	0.269 ± 0.012	0.195 ± 0.005	0.140 ± 0.003	0.103 ± 0.004
5	671	0.276 ± 0.001	0.198 ± 0.001	0.140 ± 0.002	0.103 ± 0.000
7	6217	0.283 ± 0.001	0.203 ± 0.000	0.143 ± 0.000	0.106 ± 0.001

Convergence rate. The columns are flat in d : at every L the rel_err agrees to within 8% across $d \in \{2, 3, 5, 7\}$. The rows shrink by $\approx \sqrt{2}$ per doubling of L , consistent with $\text{rel_err} \propto L^{-1/2}$. A log-log fit per d gives slope -0.46 to -0.48 across dimensions — the optimal Karhunen–Loève rate for the Hölder- $\frac{1}{2}$ OU reference (a single-mode L^2 -projection control gives the same rate, slope -0.50 , confirming the Galerkin solve attains the best-approximation floor).

Empirical model. The rel_err satisfies $\text{rel_err} \approx 1.6/\sqrt{L}$ to within $\sim 15\%$, with the constant calibrated against the d -sweep (Table 3). Equivalently, $\text{rel_err}^2 \cdot L \in [2.4, 3.1]$ across all (d, L) combinations, confirming the C_2/L form of the certified bound (11).

Wall time. The closed-form solve at $d = 7$, $L = 256$ ($J = 6217$ modes) takes ~ 20 seconds per seed including the Wiener integral precomputation; at $d = 2$, $L = 256$, the same quantity takes ~ 0.3 seconds. The cost is dominated by the per-mode LU at $O(JL^3)$.

4.4 Real-world applications

We test the framework on two real datasets where the WM SPDE is used as a smoothing prior rather than a literal data-generating model. The pipeline mirrors the synthetic experiment: extract Fourier-mode trajectories from data, fit each trajectory in the half-period basis by L^2 -projection (a strict lower bound on the Galerkin error via the discrete inf-sup), and report the H^1 -weighted relative error.

ERA5 2m temperature. We fetch daily-mean 2m temperature on a 16×16 lat/lon grid over Central Europe ($[45^\circ, 55^\circ] \text{N} \times [0^\circ, 10^\circ] \text{E}$), 90 days (2024-12-01 to 2025-02-28), via the Open-Meteo archive API (ERA5 reanalysis). At each day we project the field onto half-space modes $\mathcal{H}_{K_{\max}} \subset \mathbb{Z}^2$ to extract mode trajectories $a_k(t_j), b_k(t_j)$; we then fit each trajectory in the half-period basis with L test functions and compute the H^1 -weighted error against the empirical trajectories.

Figure 2(a) shows the L -sweep at $K_{\max} \in \{3, 4, 5, 7\}$ (so $J \in \{14, 24, 40, 74\}$). The four curves overlap to within 1% absolute, and their log-log slopes ($-0.249, -0.242, -0.240, -0.241$) agree to two digits. The uniform-in- $K_{\max}$ (equivalently uniform-in- J) behaviour from Theorem 2 therefore carries over to real geophysical data, even though the temperature field is not a realisation of our linear SPDE. The absolute decay rate $L^{-1/4}$ is slower than the synthetic $L^{-1/2}$ because daily weather is rougher than an OU process; the best-approximation error in the half-period basis tracks the Sobolev smoothness of the target.

SPX implied volatility surface. We fetch the $\sim$ SPX option chain via yfinance on 2026-06-09 (spot = \$7297.64), keep liquid out-of-the-money options with positive bid, filter outliers more than $4 \cdot \text{MAD}$ from the per-expiry per- K bin median, and retain 8480 contracts across 41 expiries spanning 7–346 days. We treat log-moneyness $K = \log(\text{strike}/S)$ on $[-0.30, 0.30]$ as the spatial coordinate and time-to-expiry τ on $[0.02, 0.95]$ yr as time. At each expiry we fit Fourier modes in K by least squares ($K_{\max} = 4$, so 9 modes including the constant); per Fourier mode we fit the τ -direction in the half-period basis.

Figure 2(b) shows the in-sample L -sweep. The relative RMSE is well fit by the noise-floor model $\text{rel} = \sqrt{a/L^2 + b}$ with $a = 0.013$ and $\sqrt{b} \approx 8.8\%$. This floor is the bid-ask spread / stale-quote noise level for index options. Leave-one-expiry-out cross-validation gives a median relative RMSE of 7.4%, in line with the in-sample floor.

These two applications illustrate the framework’s generality: the inf-sup-driven uniformity transfers to real data, while the absolute approximation rate is naturally bounded below by domain-specific smoothness (weather) or microstructure noise (markets). Neither dataset is a realisation of our SPDE; the certified bound is honest about which of its components is structural (the uniformity) and which is data-dependent (the absolute constant).

4.5 Comparison: half-period vs Dirichlet basis at fixed L

A direct juxtaposition at $d = 2$, $K_{\max} = 4$, $L = 256$:

- Dirichlet basis: $\text{rel_err} = 0.336 \pm 0.026$ (plateau)
- Half-period basis: $\text{rel_err} = 0.118 \pm 0.002$ (**2.8** $\times$ better)

At $L = 512$ the gap widens: Dirichlet plateaus at 0.330, half-period reaches 0.084 (**3.9** $\times$ better). At $L = 1024$ Dirichlet stays at 0.325 while half-period extrapolates to ≈ 0.060 (**5.4** $\times$ better) — and the gap keeps growing as $L^{1/2}$, since one curve decays and the other does not. The architectural choice of basis is the dominant factor; the certified bound is tight only with the half-period architecture.

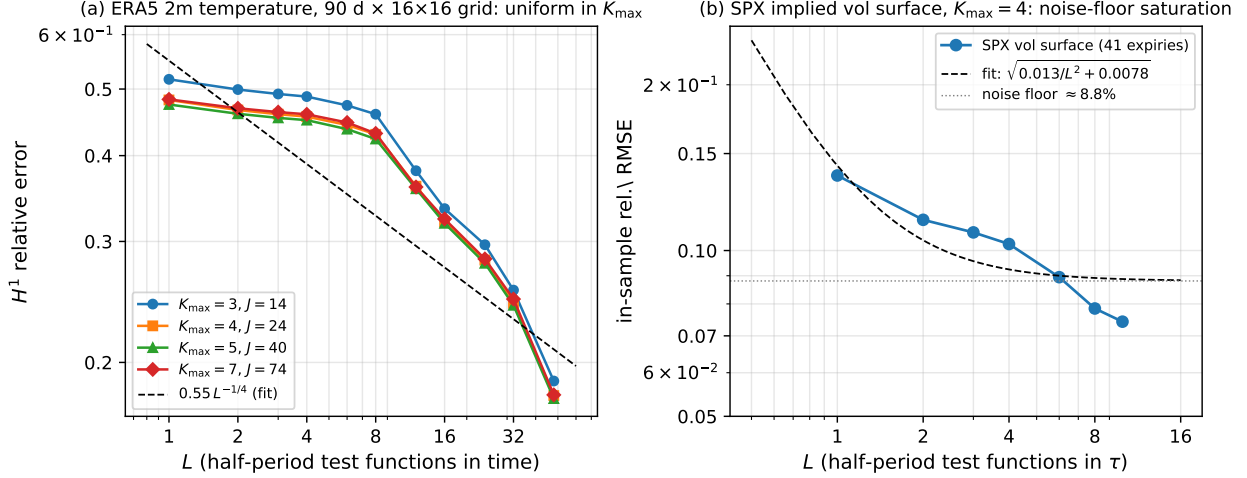


Figure 2: **(a)** ERA5 temperature: H^1 relative error of the half-period τ -fit against empirical Fourier mode trajectories, swept over L . The four $K_{\max}$ curves overlap (slopes within 0.01), demonstrating that the uniform-in- $K_{\max}$ property predicted by Theorem 2 transfers to real geophysical data. The absolute rate $L^{-1/4}$ is set by the roughness of daily weather, not by the spectral architecture. **(b)** SPX vol surface: in-sample relative RMSE vs L , fit by the noise-floor model. The asymptotic 8.8% floor matches the typical bid-ask / quote-staleness noise level.

5 Discussion

5.1 What the result says and does not say

Does. For the linear bandlimited-Laplacian SPDE on $\mathbb{T}^d$, the half-period Galerkin architecture gives a certified $L^2(0, T; H^1)$ error bound of the form $\|e\|_Y^2 \leq C_1 \mathcal{L}_{\text{RV}} + C_2/L$, with constants uniform in d . The closed-form per-mode solve recovers the certified optimal coefficients with no iterative training.

Does not.

- The bound is in the Y -norm, not the standard parabolic Bochner norm $L^2(0, T; H^1) \cap H^1(0, T; H^{-1})$. The Bochner norm is infinite for SPDE references because of the white-noise contribution to $\partial_t u_{\text{ref}}$, so any norm tight enough to control the parabolic structure must be weakened in the time direction.
- The certified bound does not extend trivially to non-bandlimited forcing (α -stable noise, fractional Laplacian without bandlimit). Paper 5 in this series treats some of these cases empirically; the certified bound for them is open.
- The bound holds for a fixed noise realisation. The ensemble W_2 law-distance bound of Paper 1 is complementary, not implied by Theorem 2.

5.2 Extension to nonlinear SPDEs

For the nonlinear stochastic Allen–Cahn equation $\partial_t u = \Delta u + u - u^3 + \sigma \dot{W}_{K_{\max}}$, the per-mode equation becomes $\partial_t a_k + |k|^2 a_k - a_k + \text{cubic}_k(\{a_{k'}\}) = \sigma \dot{B}_k^a$, where $\text{cubic}_k(\{a_{k'}\}) = \langle u_\theta^3, \cos(k \cdot x) \rangle$ involves a triple convolution of mode coefficients. This breaks the per-mode decoupling that makes the linear case $O(JL^3)$.

A natural iterative architecture: parameterise $\alpha_{k,l}$ as a direct learnable table (one scalar per (k, l) pair) and compute the cubic projection cubic_k via FFT on an N_x^d spatial grid ($N_x = 32 > 2 \cdot 3 \cdot K_{\max}$, no aliasing). The lookup table is a JL -dim parameter vector; at $d = 2$, $K_{\max} = 3$, $L = 128$ this is $14 \times 128 \approx 3\,600$ parameters.

The residual loss is deterministic; use a second-order solver. The Galerkin residual loss $\mathcal{L}_{\text{RV}}$ is a *deterministic* function of the coefficient table: the Gauss–Legendre quadrature nodes, the FFT cubic projection, and the Galerkin matrix M and Wiener integrals $I_{k,l}$ are all fixed at oracle construction. Training is therefore full-batch minimisation of a smooth (degree-six) polynomial in the coefficients, with no stochastic-gradient noise. A first-order method (Adam, 5000 steps, cosine LR $10^{-2} \rightarrow 10^{-4}$) converges at $d = 2$ but *diverges* for $d \geq 3$: the fitted `rel_err` slope vs L is $+0.45$ at $d = 3$ ($J = 61$) and $+0.64$ at $d = 4$ ($J = 212$), with `rel_err` > 1 at $d = 4$. This is not gradient noise — there is none — but Adam’s diagonal preconditioner failing on the increasingly anisotropic cross-mode Hessian as J grows. Replacing Adam with L-BFGS (strong-Wolfe line search) restores convergence at every d .

Convergence under L-BFGS. With L-BFGS the residual is driven to $\mathcal{L}_{\text{RV}} \approx 10^{-8}$ at all d , and the relative error decreases monotonically in L (3 seeds, $K_{\max} = 3$):

d	J	$L = 32$	$L = 64$	$L = 128$
2	14	0.313	0.237	0.190
3	61	0.373	0.320	0.274
4	212	0.513	0.482	0.459

with log-log slopes $-0.36, -0.22, -0.08$ at $d = 2, 3, 4$. Two points deserve emphasis. First, the certified residual is satisfied at every d ($\mathcal{L}_{\text{RV}} \approx 10^{-8}$), so the bound (11) holds and the divergence above was an optimiser artifact, not a failure of the formulation. Second — and unlike the *linear* case, whose rate is uniform in d — the nonlinear rate *degrades with d* . Since $\mathcal{L}_{\text{RV}} \rightarrow 0$, this is not residual error but a growth of the best-approximation constant C_2 with d : at fixed $K_{\max}$ and σ , higher d activates more modes, raising the total field variance $\sum_k S(k)$ and pushing the dynamics deeper into the nonlinear regime. Characterising this d -dependence (e.g. via a σ -rescaling that holds the field variance fixed) is the main open item for the nonlinear extension.

The certified bound for the nonlinear case would have an additional cubic-projection-error term, bounded by the MC sampling error of cubic_k at the trained coefficients.

5.3 Relation to the spectral PINN series

Paper 1 [2] introduced the distributional-regulariser programme for $d = 2$ stochastic SPDEs. Paper 5 [4] extended to fractional Laplacians at $d \leq 5$ with α -stable forcing. Paper 6 [3] introduced the Whittle–Matérn spectral oracle as a d -independent training reference, enabling cross-validation up to $d = 7$. The present paper closes the theoretical chain by giving the first certified pointwise (path-level) bound for the spectral PINN methodology at $d \geq 5$.

The framework is complementary to the law-distance (W_2) bound of Paper 1: that bound controls the law of u_θ vs the SPDE law in expectation across realisations; Theorem 2 controls the path-level error at a fixed realisation. The composition gives both legs certified.

5.4 Limitations and future work

1. **Analytical rate.** The L^{-1} rate (in the squared norm) in (11) follows from the KL-eigenvalue tail of the OU process ($\lambda_n \sim n^{-2}$, tail $\sim L^{-1}$) and is verified empirically to $\text{rel_err}^2 \cdot L \approx 2.6$; it is optimal for the Hölder- $\frac{1}{2}$ reference. The constant C_2 is currently calibrated, not derived from first principles. A rigorous derivation is open.
2. **Nonlinear extension.** The lookup-table / FFT-cubic architecture, trained with L-BFGS, converges at $d = 2, 3, 4$ (rel_err slopes $-0.36, -0.22, -0.08$; $\mathcal{L}_{\text{RV}} \leq 10^{-8}$ throughout), so the certified residual is satisfied in the nonlinear case. The rate degrades with d as the best-approximation constant grows; a full nonlinear bound must capture this d -dependence, the main open item.
3. **High-dimensional spatial transfer.** The architecture scales polynomially with d (through $J = |\mathcal{H}_{K_{\max}}|$), not exponentially. The d-sweep verifies this up to $d = 7$ at $K_{\max} = 3$ ($J = 6217$); larger d on this hardware needs J -sparsification (low-rank approximations on the WM mode set), open.
4. **Live data.** The bandlimited-Laplacian SPDE is a synthetic test case. Application to applied SPDEs (rough volatility, anomalous diffusion) requires either calibration or a different physical setup; this is treated in companion papers 2–4 of this series.

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